Vitamins and Minerals

STRUCTURE, FUNCTION, CLASSIFICATION, CLINICAL IMPORTANCE

LEARNING OBJECTIVES

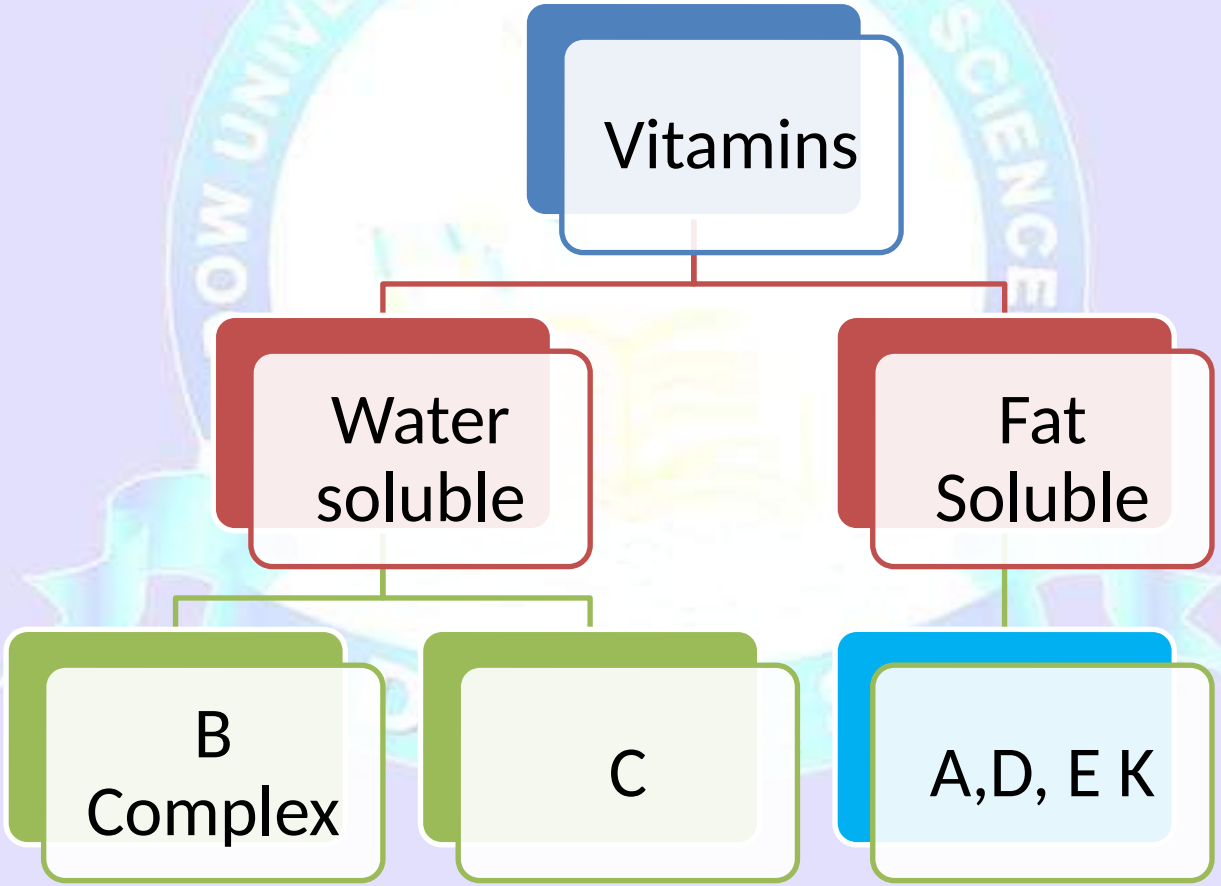
- At the end of the lecture the student should;

  - Identify vitamins as micronutrients
  - Classify vitamins on basis of solubility
  - Describe general characteristics of vitamins in relation to sources, RDA, digestion & absorption, transport in blood, storage & excretion, function, toxicity, deficiency states.
  - Define micronutrients
  - Classify minerals as macro and micro-minerals
  - Describe the sources and function of macro and micro-minerals

Vitamins

Organic compounds needed in small amount for vital functions of the body

Classified on the basis of solubility



CLASSIFICATION OF VITAMINS

WATER SOLUBLE

- B complex
  - Thiamin
  - Riboflavin
  - Niacin
  - Biotin
  - Pantothenic Acid
  - Vitamin B-6
  - Folate
  - Vitamin B-12
- Vitamin C

FAT SOLUBLE

- Vitamin A
- Vitamin D
- Vitamin E
- Vitamin K

General Characteristics of Vitamins

Characteristics	Water soluble	Odd one out	Fat soluble	Odd one out
Sources	Plant sources Green leafy vegetables, Fruits	B12 in liver	Animal sources Egg, Lean and Organ meat, Fish, Oils Plant sources (A, E)	K also found in plants D synthesized by cholesterol in skin
Digestion & Absorption	Like water soluble substances, in portal blood	B12 needs intrinsic factor	Through micelles into lacteales	-
Transport in blood	Free dissolved in plasma	B12 with a carrier Trans-cobalamine	In lipoproteins	-
Storage	Not stored	B12 stored in Liver	In Liver	-
Excretion	In Urine	-	In Bile, through Feces	-
RDA	Large amounts, in mg	B12	Low amounts, in µg	-

OVERVIEW OF VITAMINS

WATER SOLUBLE

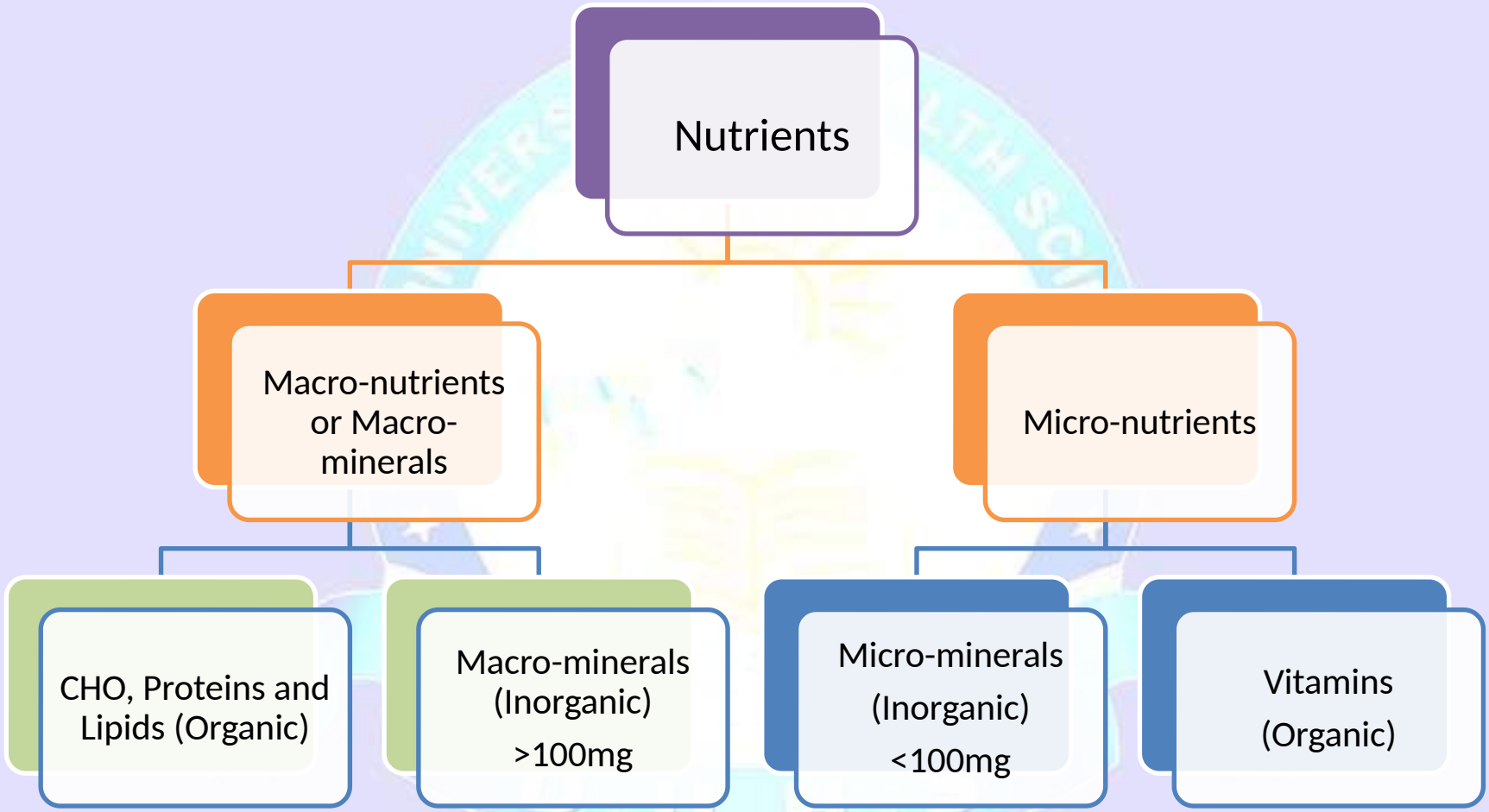
- Widely spread in nature
- Function as a coenzyme & in energy metabolism
- Readily excreted by kidney
- Vitamin C, thiamin and riboflavin are especially susceptible to heat and alkalinity
- Marginal deficiency states are more common

OVERVIEW OF VITAMINS

FAT SOLUBLE

- Adsorbed with dietary fat in micelles
- Chylomicrons containing fat-soluble vitamins are transported via the lymph to the bloodstream and evetually to the liver.
- Stored in adipose tissue & liver
- Excreted much more slowly
- Greater risk of toxicity when consumed in excess

Minerals


Nutrients
Macro-nutrients or Macro-minerals
Micro-nutrients
CHO, Proteins and Lipids (Organic)
Macro-minerals (Inorganic) >100mg
Micro-minerals (Inorganic) <100mg
Vitamins (Organic)

Minerals

Macr-ominerals

- Sodium
- Potassium
- Chloride
- Magnesium
- Calcium
- Phosphate
- Sulphur

Micro-minerals

- Iron
- Copper
- Cobalt
- Zinc
- Manganese
- Fluoride

Functions of Minerals

Function	Minerals	Processes
Structural	Calcium, Magnesium, Phosphorus,	Muscle and Bone structure, Muscle contractions
Membrane structure and function	Sodium, Potassium, Calcium	Membrane potential, Action Potential, Na+, K+ ATPase
Prosthetic Group	Iron, Cobalt, Copper, Selenium, Zinc	Hemoglobin, Myoglobin, Enzyme activities
Hormone action and release	Calcium, Iodine, Sodium, Potassium	Thyroxin, hormone secretions, catecholamine
Enamel and dentine health	Flouride	Prevent carries

Macro-minerals

Mineral	Functions	Sources
Calcium	Bones and teeth metabolism, Muscle contraction, Nerve transmission, hormone release, cardiac muscle functions	Milk, Dairy products, Almonds, Spinach
Chloride	Main anion of plasma, nutrient metabolisms, CO_2 transport, production of glandular secretions, sodium and water homeostasis	Table salt, dairy products, olives
Sodium	Main cation of plasma, water homeostasis, Blood pressure regulation, Action potential, membrane potential	Table salt, dairy products, Meat, Fish, poultry

Macro-minerals

Mineral	Functions	Sources
Potassium	Main anion of intracellular fluid, Action potential, Membrane potential, cardiac muscle functions	Fruits, banana, Apricot, spinach, vegetables
Phosphorus	Meat, poultry, dairy products	Calcium homeostasis, bones and teeth metabolism, DNA, RNA structure, Energy carrier, 2nd messenger systems
Magnesium	Almond, beans, spinach, pumpkins	Bone and macronutrient metabolism
Sulphur	GAGs, ECM, Hair, Nail, and skin	Vegetables, Dairy products, eggs

Micro-minerals

Mineral	Functions	Sources
Iron	Heme and Non-heme, Hemoglobin, myoglobin, cytochromes, ETC and OP, Catalase,	Dairy products, Banana, Guava, Spinach, Coloured vegetables
Copper	Antioxidant, role in synthesis and stabilization of fibrous proteins	Fish, Organ meat, Dry fruits,
Iodine	Thyroid hormones	Fish, strawberries
Manganese	Metabolism of macronutrient, help maintain blood glucose levels	Vegetables, pineapple, Spinach, Black tea
Zinc	Antioxidant, Role in spermatogenesis, hormone function and gene expression	Meat, Dairy products, Cheese
Nickle	Role in certain enzyme functions	Legumes, dry fruits
Cobalt	Part of vitamin B12	Organ meat, lean meat, leafy begetables
Boron	Stabilizes Ca^{+2} and PO_4^{-2} metabolism	Citreous fruits and dry fruits

References

- Harpers Biochemistry 30th edition
  - Chapter 44: Micronutrients, Vitamins and Minerals
- Lippincott Biochemistry 5th edition
  - Chapter 28: Vitamins